

Mohamed Amine Hajji

AI/ML & Software Engineer

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Summary

Software Engineer specializing in **applied AI**, **optimization**, and **full-stack systems**. Experience building production **ML pipelines**, **scheduling engines**, **geospatial platforms**, and data-intensive applications using **Python**, **TypeScript**, **Angular**, **PostgreSQL**, and **AWS**. Strong algorithmic problem-solving background with a focus on turning complex technical problems into scalable production solutions.

Experience

Software & AI Engineering Intern (End-of-Studies Project), Drivago – Tunisia Feb 2026 – June 2026

- Designed and built a **machine learning pipeline** to predict candidate driving-lesson hours from historical candidate data.
- Designed and developed a **7-step digital candidate assessment application** with **Angular** and a **Three.js** 3D cockpit simulator, feeding candidate readiness scores into the hours-prediction model to calibrate its estimates.
- Designed and built an **optimization-based scheduling engine** (**Python**, **FastAPI**, **Google OR-Tools CP-SAT**) exposed as a **REST API** consumed by the Node.js backend, automatically generating lesson schedules — solving a full 3-week schedule for **50 students** and **12 instructors** in about **1.2 seconds** with **100% of sessions successfully placed**.
- Trained a **churn-prediction model** (**0.875 ROC-AUC** — correctly ranks at-risk customers 87.5% of the time) to flag them before they cancel their subscription.
- Built **Looker Studio** dashboards on **PostgreSQL** data covering **600+ partner driving schools**, giving the product team visibility into revenue, feature adoption, and churn signals.
- Tracked and debugged performance issues in an **Angular** application using **Chrome DevTools** and stack traces, reducing interaction latency by **75%** on average and **87%** in the worst case.

Full-Stack Software Engineer - Part-Time, DeepVolt – Ariana, Tunisia July 2023 – Jan 2026

- Migrated the platform's heatmap rendering from **GeoJSON** to a **vector-tile pipeline** (using **Tippecanoe**, hosted on **AWS S3**), cutting load times by **75-85%** (low-zoom: 8-12s to 1-2s).
- Led the adoption of **Martin**, an open-source **Rust**-based **PostGIS** tile server: wrote the PostGIS queries and deployed it to add real-time filtering and clustering across heatmap layers.
- Built configurable competitor, POI, and traffic filters — designed the **REST API** endpoints (**Node.js/PostgreSQL** backend) and the **Angular** frontend — letting users fine-tune how the site-scoring algorithm weighs nearby competition, points of interest, and road traffic.
- Optimized data fetching from the **TomTom API** and integrated a purchased dataset of **100,000+ EV charging stations** across France into the platform's map layers.
- Supported the data science team with data cleaning and web scraping to build training datasets, using **QGIS** to visualize and validate geospatial data.
- Worked on the front-end of a side project for DeepVolt's **Istanbul** expansion, integrating the team's AI-recommended charging-station locations and connecting the UI to their backend endpoints.

Front-End Developer Intern (End-of-Studies Project), DeepVolt – Ariana, Tunisia Feb 2023 – June 2023

- Built the front-end of **DLIA** (DeepVolt Location Intelligence Assistant), a tool that helps identify the best locations for EV charging stations, including **JWT-based authentication** and an admin panel for managing users and companies.
- Built the map exploration experience: point-of-interest markers powered by the **TomTom API**, a location details panel, and toggleable heatmap layers for area potential score, population density, income, and EV registrations.

- Built the **multi-location comparison** feature, letting users compare several candidate sites side by side, and integrated the **consumption-prediction** flow into the UI so users could predict a site's expected EV charging demand.

Projects

Gender Classification on the Adience Dataset

Deep learning model for gender classification from facial images, built with a teammate for an advanced AI course.

- Built a face-detection preprocessing pipeline (**OpenCV** Haar Cascade) and fine-tuned **EfficientNet-B0** via transfer learning in **PyTorch**.
- Reached **96% test accuracy** on 3,247 held-out images from a 16K-image training set.

Research Portal Web Application

A full-stack web app for sharing and exploring research papers, with role-based access and an integrated research chatbot.

- Built with **Angular**, **Node.js/Express**, and **PostgreSQL**, with role-based access for students, researchers, and general users; integrated a **Hugging Face**-powered chatbot for contextual Q&A on each paper.

Education

Higher Institute of Computer Science (ISI), National Engineering Degree in Software Engineering – Ariana, Tunisia Sept 2023 – June 2026

Higher Institute of Computer Science (ISI), Bachelor's Degree in Computer Science – Ariana, Tunisia Sept 2020 – June 2023

Borj Louzir High School, Baccalaureate (Honors) in Mathematics – Tunisia Sept 2016 – June 2020

Skills

Languages: Python, TypeScript, JavaScript, SQL, C, C++, Rust

Frameworks & Tools: Angular, Node.js, Express, FastAPI, Three.js, Git

Geospatial & Mapping: Mapbox, TomTom API, Martin, Tippecanoe, QGIS

Databases: PostgreSQL, PostGIS, MySQL, MongoDB

Machine Learning & Data: PyTorch, Scikit-learn, Google OR-Tools (CP-SAT), Looker Studio

Cloud & DevOps: AWS (EC2, RDS, S3), Docker, GitHub Actions (CI/CD)

Achievements

- Meta Hacker Cup 2022 — Reached Round 2 (Global Rank: 2993) — [Certificate](#)
- CodinGame Top 0.6% Global Ranking — [Profile](#)

Volunteer

- Freeways ISI — Problem Solving Department

Languages

Arabic (Native), French (Intermediate), English (Fluent)